



第四节 变系数方程

变系数初值问题:

$$\begin{cases} \frac{\partial u}{\partial t} - a(x) \frac{\partial^2 u}{\partial x^2} = 0(f(x, t)) \\ u(x, 0) = g(x) \end{cases},$$

$$a(x) \geq a_0 > 0, \quad a(x) \text{连续}$$

方程两端在节点 (j, n) 取值, 即

$$\begin{cases} \left. \frac{\partial u}{\partial t} \right|_j^n - a(x)|_j \left. \frac{\partial^2 u}{\partial x^2} \right|_j^n = 0(f(x, t)|_j^n) \\ u(x|_j, 0) = g(x|_j) \end{cases}$$

$$a(x) \geq a_0 > 0$$



4.1、Taylor级数展开法

由于

$$\begin{aligned}\left[\frac{\partial^2 u}{\partial x^2}\right]_j^n &= \frac{u(x_{j+1}, t_n) - 2u(x_j, t_n) + u(x_{j-1}, t_n)}{h^2} + O(h^2) \\ &= \left[\frac{1}{a(x)} \frac{\partial u}{\partial t}\right]_j^n + O(h^2)\end{aligned}$$



引入时间变量差商，得近似方程的差分格式：

$$\frac{u_j^{n+1} - u_j^n}{\tau} = a_j \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{h^2}$$

古典显式格式，其截断误差为 $E = O(\tau + h^2)$

稳定性条件(冻结系数法) $\max_x |a(x)| \lambda \leq \frac{1}{2}$

$$\frac{u_j^{n+1} - u_j^n}{\tau} = a_j \frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{h^2}$$

古典隐式格式，其截断误差为 $E = O(\tau + h^2)$

稳定性条件(冻结系数法)：无条件稳定

若 $a(x)$ 高阶连续可导，可构造更高精度的差分格式，

$$\text{设 } \delta_x^2 u_j^n = u(x_{j+1}, t_n) - 2u(x_j, t_n) + u(x_{j-1}, t_n)$$

那么：

$$\begin{aligned} \frac{\delta_x^2 u_j^n}{h^2} &= \left[\frac{\partial^2 u}{\partial x^2} \right]_j^n + \frac{h^2}{12} \left[\frac{\partial^4 u}{\partial x^4} \right]_j^n + O(h^4) \\ &= \left(\frac{1}{a(x)} \frac{\partial u}{\partial t} \right)_j^n + \frac{h^2}{12} \left(\frac{\partial^2}{\partial x^2} \left(\frac{1}{a(x)} \frac{\partial u}{\partial t} \right) \right)_j^n + O(h^4) \end{aligned}$$



$$= \left(\frac{1}{a(x)} \frac{\partial u}{\partial t} \right)_j^n + \frac{h^2}{12} \left\{ \frac{\delta_x^2 \left(\frac{1}{a(x)} \frac{\partial u}{\partial t} \right)}{h^2} \Big|_j^n + O(h^2) \right\} + O(h^4)$$

$$= \left(\frac{1}{a(x)} \frac{\partial u}{\partial t} \right)_j^n + \frac{h^2}{12} \left\{ \frac{\delta_x^2 \left(\frac{1}{a(x)} \frac{\partial u}{\partial t} \right)}{h^2} \Big|_j^n + O(h^2) \right\} + O(h^4)$$

$$= \frac{5}{6} \left(\frac{1}{a(x)} \frac{\partial u}{\partial t} \right)_j^n + \frac{1}{12} \left(\frac{1}{a(x)} \frac{\partial u}{\partial t} \right)_{j+1}^n + \frac{1}{12} \left(\frac{1}{a(x)} \frac{\partial u}{\partial t} \right)_{j-1}^n + O(h^4)$$

引入时间差商，得差分格式：

$$\frac{1}{12} \frac{u_{j+1}^{n+1} - u_{j+1}^n}{a_{j+1} \cdot \tau} + \frac{5}{6} \frac{u_j^{n+1} - u_j^n}{a_j \cdot \tau} + \frac{1}{12} \frac{u_{j-1}^{n+1} - u_{j-1}^n}{a_{j-1} \cdot \tau} = \frac{\delta_x^2 u_j^{n+1} + \delta_x^2 u_j^n}{2h^2}$$

截断误差 $E = O(\tau^2 + h^4)$ ，即为紧格式。

稳定性条件(冻结系数法)：无条件稳定



4.2、Keller盒式格式

1970年, *H.B Keller*提出了解抛物型初边值问题的一个盒式格式, 这一格式具有二阶精度, 且无条件稳定。

初边值问题 :

$$\begin{cases} \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(a(x) \frac{\partial u}{\partial x} \right) & a(x) \geq a_0 > 0 \\ u(x, 0) = g(x) & 0 < x < 1 \\ u(0, t) = g_0(x) & t < 0 \\ u(1, t) = g_1(x) & \end{cases}$$


首先将方程化为一阶微分方程组：

$$\begin{cases} a(x) \frac{\partial u}{\partial x} = v(x, t) & (x, t) \in \{(x, t) | 0 \leq x \leq 1, 0 \leq t \leq T\} \\ \frac{\partial v}{\partial x} = \frac{\partial u}{\partial t} \end{cases}$$

在定解区域上取网格：

$$R_h(T) = \left\{ (x_j, t_n) \left| \begin{array}{l} x_j = jh, j = 0, 1, \dots, J, Jh = 1 \\ t_n = n\tau, n = 0, 1, \dots, N, N\tau = T \end{array} \right. \right\}$$

引入记号：



$$(1) x_{j \pm \frac{1}{2}} = \frac{1}{2} (x_j + x_{j \pm 1}), t_{n \pm \frac{1}{2}} = \frac{1}{2} (t_n + t_{n \pm 1})$$

$$(2) \phi_j^{n \pm \frac{1}{2}} = \frac{1}{2} (\phi_j^n + \phi_j^{n \pm 1}), \phi_{j \pm \frac{1}{2}}^n = \frac{1}{2} (\phi_j^n + \phi_{j \pm 1}^n)$$

这样 ϕ_j^n 是定义在 $R_h(T)$ 上的函数，在 $R_h(T)$ 上使用中心差分近似方程组：

$$\left\{ \begin{array}{l} a_{j-\frac{1}{2}} \frac{u_j^n - u_{j-1}^n}{h} = v_{j-\frac{1}{2}}^n \\ \frac{v_j^{n-\frac{1}{2}} - v_{j-1}^{n-\frac{1}{2}}}{h} = \frac{u_{j-\frac{1}{2}}^n - u_{j-\frac{1}{2}}^{n-1}}{\tau} \end{array} \right.$$

$$j = 1, 2, \dots, J; n = 1, 2, \dots, N.$$

初始条件的离散

$$\begin{cases} u_j^0 = g(x_j) \\ v_j^0 = a_j \frac{dg(x_j)}{dx} \end{cases} \quad j = 1, 2, \dots, J$$

边界条件离散:

$$\begin{cases} u_0^n = g_0(t_n) \\ u_1^n = g_1(t_n) \end{cases} \quad 1 \leq n \leq N$$

格式特点:

截断误差: 二阶精度

稳定性条件: 无条件稳定

容易编制程序

可以应用到非线性问题及间断系数问题



4.3、有限体积法

考虑方程：

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(a(x) \frac{\partial u}{\partial x} \right), \quad a(x) \geq a_0 > 0$$

令 $w(x, t) = a(x) \frac{\partial u}{\partial x}$ ，那么方程变成 $\frac{\partial u}{\partial t} = \frac{\partial w}{\partial x}$

对上面的方程在 $[x_{j-\frac{1}{2}}, x_{j+\frac{1}{2}}] \times [t_n, t_{n+1}]$ 上积分有

$$\int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} [u(x, t_{n+1}) - u(x, t_n)] dx$$

$$= \int_{t_n}^{t_{n+1}} [w(t, x_{j+\frac{1}{2}}) - w(t, x_{j-\frac{1}{2}})] dt$$



据 $w(x, t)$ 的定义 有 $\frac{\partial u}{\partial x} = \frac{w(x, t)}{a(x)}$, 在 $[x_j, x_{j+1}]$ 上积分有

$$u(x_{j+1}, t) - u(x_j, t) = \int_{x_j}^{x_{j+1}} \frac{w(x, t)}{a(x)} dx$$

$$\approx w(x_{j+\frac{1}{2}}, t) \int_{x_j}^{x_{j+1}} \frac{1}{a(x)} dx$$

$$w(x_{j+\frac{1}{2}}, t) \approx \frac{h}{\int_{x_j}^{x_{j+1}} \frac{1}{a(x)} dx} \bullet \frac{u(x_{j+1}, t) - u(x_j, t)}{h}$$

令
$$A_{j+1} = h \left(\int_{x_j}^{x_{j+1}} \frac{1}{a(x)} dx \right)^{-1}$$

则有

$$w(x_{j+\frac{1}{2}}, t) \approx A_{j+1} \bullet \frac{u(x_{j+1}, t) - u(x_j, t)}{h}$$

积分近似：

$$\int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} u(x, t_n) dx \approx hu(x_j, t_n),$$

$$\int_{t_n}^{t_{n+1}} w(t, x_{j+\frac{1}{2}}) dt \approx \tau [\theta w(t_{n+1}, x_{j+\frac{1}{2}}) + (1-\theta)w(t_n, x_{j+\frac{1}{2}})]$$

于是有

$$\frac{u(x_j, t_n) - u(x_j, t_{n-1})}{\tau}$$

$$= \frac{1}{h^2} [\theta \Delta(A_j \nabla u(x, t_{n+1})) + (1 - \theta) \Delta(A_j \nabla u(x, t_n))] + R_j^n$$

Δ 、 ∇ 分别表示向前和向后差分, 即

$$\Delta u(x_j, t_n) = u(x_{j+1}, t_n) - u(x_j, t_n)$$

$$\nabla u(x_j, t_n) = u(x_j, t_n) - u(x_{j-1}, t_n)$$

忽略误差项, 得差分格式

$$\frac{u_j^{n+1} - u_j^n}{\tau} - \frac{1}{h^2} [\theta \Delta(A_j \nabla u_j^{n+1}) + (1 - \theta) \Delta(A_j \nabla u_j^n)] = 0$$

A_j 的计算：

- 1、数值积分的误差阶不低于差分格式的截断误差。
- 2、一般可取为

$$A_j = a(x_{j-\frac{1}{2}}) \quad \text{或} \quad A_j = \frac{1}{2} [a(x_{j-1}) + a(x_j)]$$

格式的特点：

1、截断误差： $\theta = \frac{1}{2}$, $R_j^n = O(\tau^2 + h^2)$

$$\theta \neq \frac{1}{2}, R_j^n = O(\tau + h^2)$$

2、稳定性 $\frac{1}{2} \leq \theta \leq 1$, 无条件稳定

$$0 \leq \theta < \frac{1}{2}, \text{条件稳定}$$

实际上： $\theta = 0$, 显式格式, $\theta = 1$, 隐式格式

$$\theta = \frac{1}{2}, \text{Crank - Nicolson格式}$$

4.4、间断系数问题

考虑方程

$$\frac{\partial u}{\partial t} - \frac{\partial}{\partial x} \left(a(x) \frac{\partial u}{\partial x} \right) + b(x) = 0, \quad a(x) \geq a_0 > 0$$

假定在 $x = \xi$ 上, $a(x), b(x)$ 有第一类间断点

在间断线上解满足如下联结条件

$$\begin{cases} u(\xi - 0, t) = u(\xi + 0, t) \\ a(\xi - 0) \frac{\partial u}{\partial x}(\xi - 0, t) = a(\xi + 0) \frac{\partial u}{\partial x}(\xi + 0, t) \end{cases}$$

为了处理问题的方便，通常使间断线与网格线重合。

即、 $x_j = \xi$

采用有限体积方法：

积分区域： $D = \{(x, t), x_{j-\frac{1}{2}} \leq x \leq x_{j+\frac{1}{2}}, t_n \leq t \leq t_{n+1}\}$

对上面的方程在 D 上积分有

$$\begin{aligned} & \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} \int_{t_n}^{t_{n+1}} \frac{\partial u(x, t)}{\partial t} dt dx \\ &= \int_{t_n}^{t_{n+1}} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} \frac{\partial}{\partial x} w(x, t) dt - \int_{t_n}^{t_{n+1}} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} b(x) u(x) dt \end{aligned}$$



$$\int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} \int_{t_n}^{t_{n+1}} \frac{\partial u(x, t)}{\partial t} dt dx = \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} [u(x, t_{n+1}) - u(x, t_n)] dx$$

$$\int_{t_n}^{t_{n+1}} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} \frac{\partial}{\partial x} w(x, t) dt$$

$$= \int_{t_n}^{t_{n+1}} \left[\int_{x_{j-\frac{1}{2}}}^{x_j} \frac{\partial}{\partial x} w(x, t) dx \int_{x_j}^{x_{j+\frac{1}{2}}} \frac{\partial}{\partial x} w(x, t) dx \right] dt$$

$$= \int_{t_n}^{t_{n+1}} [w(x_{j+\frac{1}{2}}, t) - w(x_j + 0, t) + w(x_j - 0, t) - w(x_{j-\frac{1}{2}}, t)] dt$$

$$= \int_{t_n}^{t_{n+1}} [w(x_{j+\frac{1}{2}}, t) - w(x_{j-\frac{1}{2}}, t)] dt$$



$$\begin{aligned}
& \int_{t_n}^{t_{n+1}} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} b(x)u(x)dt \\
&= \int_{t_n}^{t_{n+1}} \left[\int_{x_{j-\frac{1}{2}}}^{x_j} b(x)u(x) + \int_{x_j}^{x_{j+\frac{1}{2}}} b(x)u(x) \right] dx dt \\
&= \int_{t_n}^{t_{n+1}} \frac{h}{2} [b(x_j + 0) + b(x_j - 0)] u(x_j, t) dt \\
&= \frac{\tau h}{2} [b(x_j + 0) + b(x_j - 0)] u(x_j, t_n)
\end{aligned}$$



可得差分格式:

$$\frac{u_j^{n+1} - u_j^n}{\tau} - \frac{1}{h^2} \Delta(A_j \nabla u_j^n) + \frac{1}{2} (b_{j+0} + b_{j-0}) u_j^n = 0$$

$$A_j = h \left(\int_{x_{j-1}}^{x_j} \frac{1}{a(x)} dx \right)^{-1}$$

注:由此看出有限体积法在处理间断系数问题的有效性.
对于右端项有间断的情形同样适用.



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